Barnaby **Seaman**

Aero-Mechanical Engineer

Kilmarnock, Ayrshire

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Personal Profile

MEng Aero-Mechanical Engineering graduate with hands-on experience in launch vehicle systems. Proven ability to contribute within interdisciplinary teams, take ownership of technical tasks, and rapidly develop new capabilities.

Skills

Engineering	**High-Power Rocketry**, Aerodynamic Analysis, Composite Structures, Additive Manufacturing
Analysis & Simulation	**FEA (ANSYS Mechanical)**, CFD (ANSYS Fluent), Scripting (Python, MATLAB), MS Office Suite
Design & Software	**CAD (SOLIDWORKS)**, Git, Typst / LaTeX
Leadership & PM	**Subsystems Interface Management**, PDR / CDR / FRR Participation, Risk Assessment, WBS & Gantt-based Planning

Experience

President — University of Strathclyde

Strathclyde Students for the Exploration and Development of Space (StrathSEDS) — August 2023 - July 2025

- Led a 20-member multidisciplinary rocketry team across mechanical, electrical, and software subteams
- Provided technical onboarding in CAD, aerodynamic analysis, and design techniques for additive manufacturing
- Delivered two national competition entries per year (Mach-X, National Rocketry Championship), achieving podium finishes in three of four events
- Planned and managed technical projects using work breakdown structures and Gantt charts to meet competition timelines and deadlines
- Oversaw system-level design through the full design review process and launch operations
- Completed tests to in-house and competition-set requirements
- Coordinated subsystem interfaces and managed integration risks across propulsion, structures, avionics, and recovery systems
- Contributed to aerodynamic and structural design of mid- and high-power rockets using composite materials and additively manufactured plastic components

Team Member — University of Strathclyde

StrathSEDS — January 2022 - July 2026

- Contributed to a 1st-place finish in the 1 km Vehicle and Payload category at Mach-22
- Designed and built launch vehicles with motors up to L-class to altitudes exceeding 3km
- Developed part of an open-loop antenna pointing system to enable real-time video reception from a near-sonic vehicle

Systems Team Member — Remote

Launch 4 Change Alumni Team — October 2026 - Present

- Challenging myself in a new field in electronic design
- Designing a climate science payload to be used in a high-power rocket

Bus Driver — Glasgow

First Glasgow (No. 1) Ltd. — September 2023 - December 2025

- Obtained Category D Passenger Carrying Vehicle licence
- Delivered customer-facing services and conducted pre-service vehicle safety and cleanliness checks

Bus Driver — Glasgow

West Coast Motors (Craig of Campbeltown) — July 2026 - Present

- Delivering customer-facing service in service buses and coaches

Education

University of Strathclyde — Glasgow

MEng Aero-Mechanical Engineering (Distinction) — September 2020 - June 2026

- Accredited by IMechE and RAeS
- Achieved Distinction (≥ 70%)
- Coursework emphasis on lightweight structures, structural analysis by hand and by computer, finite element methods, and CFD

MCA[6] — Mildenhall, Suffolk

A-Levels — September 2018 - July 2020

- OCR Further Maths - A*
- OCR Maths - A*
- AQA Physics - A*
- OCR Computer Science - A

Iceni Academy — Methwold, Norfolk

GCSEs at grades 8-5 — September 2013 - July 2018

Projects

Airbus A380 Bearing Bracket Lightweighting — bsmn.space/ME533

University Coursework — January - April 2026

- Conducted topology optimisation on a bearing bracket, reducing its mass by 49% while maintaining a factor of safety of 1.36
- Modelled various phases of the lightweight part in ANSYS Mechanical
- Recreated optimised part in SOLIDWORKS with fabrication requirements in mind
- Completed mesh convergence studies on various stages of optimisation for verification

Solar Sail–Based Satellite Deorbiting Study — barnaby2002/sailing-home

Undergraduate Dissertation — September 2024 - May 2025

- Analysed the feasibility of solar sails for end-of-life deorbiting of defunct satellites
- Modelled orbital decay behaviour under continuous low-thrust solar radiation pressure
- Evaluated the influence of sail area and orbital parameters on deorbit times in LEO and MEO

Trans-Atmospheric Flight Simulation Enhancements — strath-ace/TITAN

Master's Group Project — September 2025 - March 2026

- Added support for control strategy handling within an atmospheric entry simulation framework
- Integrated control inputs into existing vehicle dynamics and trajectory models
- Contributed to software development within a collaborative group project

High-Power Certification — United Kingdom Rocketry Association

Independent Certification Project — July 2022 - Present

- Designed multiple high-power rocket concepts using commercial solid rocket motors
- Analysed stability margins, mass properties, and recovery system sizing for safe flight
- Produced CAD models and design documentation in preparation for future construction and flight
- Completed first flight of three for Level 1 certification

Continuous Professional Development

- Working towards IMechE Chartership

Certifications

04/05/2023 — Certified SOLIDWORKS Associate (CSWA)

20/05/2024 — Certified SOLIDWORKS Professional (CSWP)

References

References available on request.